ERP Solution for Digital Enterprises: SD, MM & PP Implementation for automobile manufacturing for industries

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A PBL report for the ERP Solution for Digital Enterprises

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INDEX / TABLE OF CONTENTS :

1. Introduction

The Automobile Manufacturing Industry is one of the most significant sectors contributing to economic growth, employment generation, and technological advancement. It involves the design, development, manufacturing, and sale of motor vehicles such as cars, trucks, and electric vehicles.

With increasing global competition and demand for high-quality products, automobile companies are rapidly adopting digital technologies. One of the key technologies is Enterprise Resource Planning (ERP), which integrates various business functions such as production, procurement, sales, inventory, and finance into a unified system.

SAP ERP plays a crucial role in automobile manufacturing by streamlining operations, improving efficiency, and enabling real-time decision-making. This report focuses on implementing SAP modules such as Sales and Distribution (SD), Material Management (MM), and Production Planning (PP) in an automobile manufacturing environment.

2. About Automobile Manufacturing Industry

The automobile manufacturing industry involves large-scale production of vehicles using advanced machinery, robotics, and automation systems.

Key characteristics include:

- High volume production with standardized processes
- Complex supply chain involving multiple vendors
- Integration of mechanical, electrical, and software components
- Continuous innovation in electric and autonomous vehicles

Major activities include:

- Vehicle design and engineering
- Procurement of raw materials and components
- Assembly line production
- Quality testing and inspection
- Sales and distribution

3. Existing ERP Implementation

The automobile industry uses SAP ERP to integrate business processes across departments.

Key details:

- SAP Version: ECC 6.0 or S/4HANA
- Users: Large workforce including production, logistics, and management
- Modules implemented:
    - FI CO (Finance and Controlling)
    - MM (Material Management)
    - SD (Sales and Distribution)
    - PP (Production Planning)
    - QM (Quality Management)
    - HCM (Human Resource Management)

ERP enables centralized data management and seamless communication between departments.

4. Key Challenges before SAP ERP was implemented

Before ERP implementation, automobile companies faced several issues:

- Lack of integration between departments
- Manual data entry leading to errors
- Poor inventory visibility
- Delays in procurement and production planning
- Difficulty in tracking orders and deliveries
- Inefficient communication across supply chain
- Limited real-time decision-making capability

5. Reasons for Choosing SAP

SAP was chosen due to the following reasons:

- Highly integrated system covering all business processes
- Scalable and suitable for large manufacturing industries
- Strong support for production and supply chain management
- Real-time data processing and reporting
- Industry best practices and standardization
- Improved transparency and control

Proposed Implementation

To enhance operational efficiency and improve user experience, automobile manufacturing companies propose the implementation of SAP Fiori. SAP Fiori provides a modern, user-friendly, and role-based interface that simplifies complex SAP processes and enables better accessibility.

Key Features:

- SAP Fiori offers a role-based user interface, allowing employees to access only the information and functions relevant to their job roles.
- It provides mobile accessibility, enabling shop floor workers and managers to use the system through smartphones and tablets.
- The system supports real-time data updates, ensuring that all users have access to the latest information for accurate decision-making.
- It simplifies workflows by reducing unnecessary steps and improving process efficiency.
- SAP Fiori enables faster approvals and decision-making by allowing managers to approve requests instantly from anywhere.

7. Business Drivers for Fiori Adoption

The adoption of SAP Fiori is driven by several business needs aimed at improving efficiency and usability.

- There is a strong need for mobile access in production units so that employees can perform tasks directly from the shop floor without relying on desktop systems.

- Organizations require a simplified user interface to reduce complexity and make the system easier for employees to use.
- Faster approval processes are essential to avoid delays in procurement, production, and sales activities.
- Improved user experience is necessary to increase employee productivity and reduce errors.
- Reduced training time is important, as a user-friendly interface helps employees learn and adapt quickly to the system.

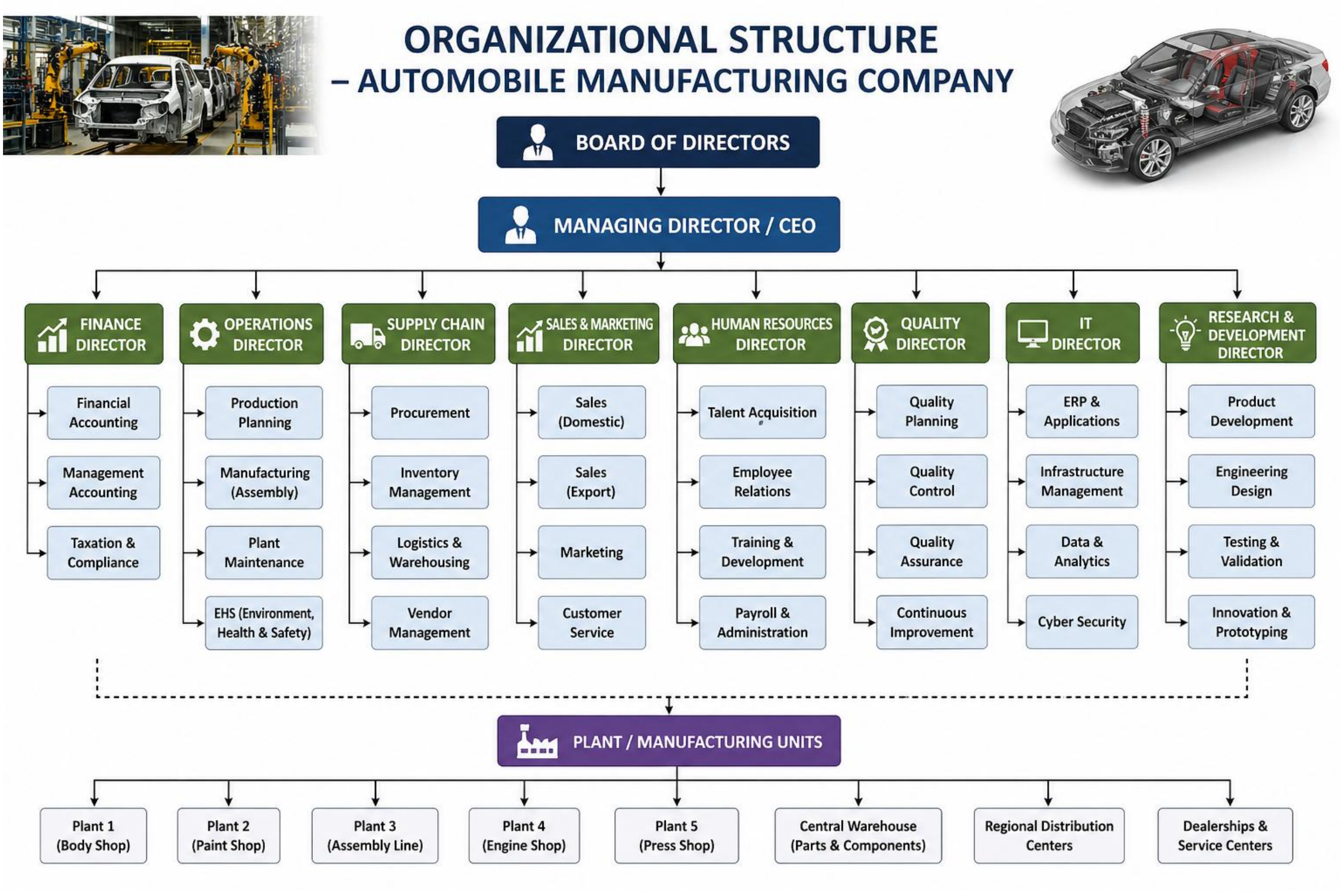
8. Expected Strategic Benefits

The implementation of SAP Fiori is expected to provide several strategic benefits to automobile manufacturing companies.

- It increases productivity by enabling employees to complete tasks faster and more efficiently.
- It improves data accuracy by allowing real-time data entry and reducing manual errors.
- It supports faster decision-making by providing instant access to updated information and analytics.
- It enhances overall operational efficiency by streamlining business processes and reducing delays.
- It improves customer satisfaction by ensuring timely delivery and better service quality.
- It reduces operational costs by minimizing manual work, errors, and process inefficiencies.

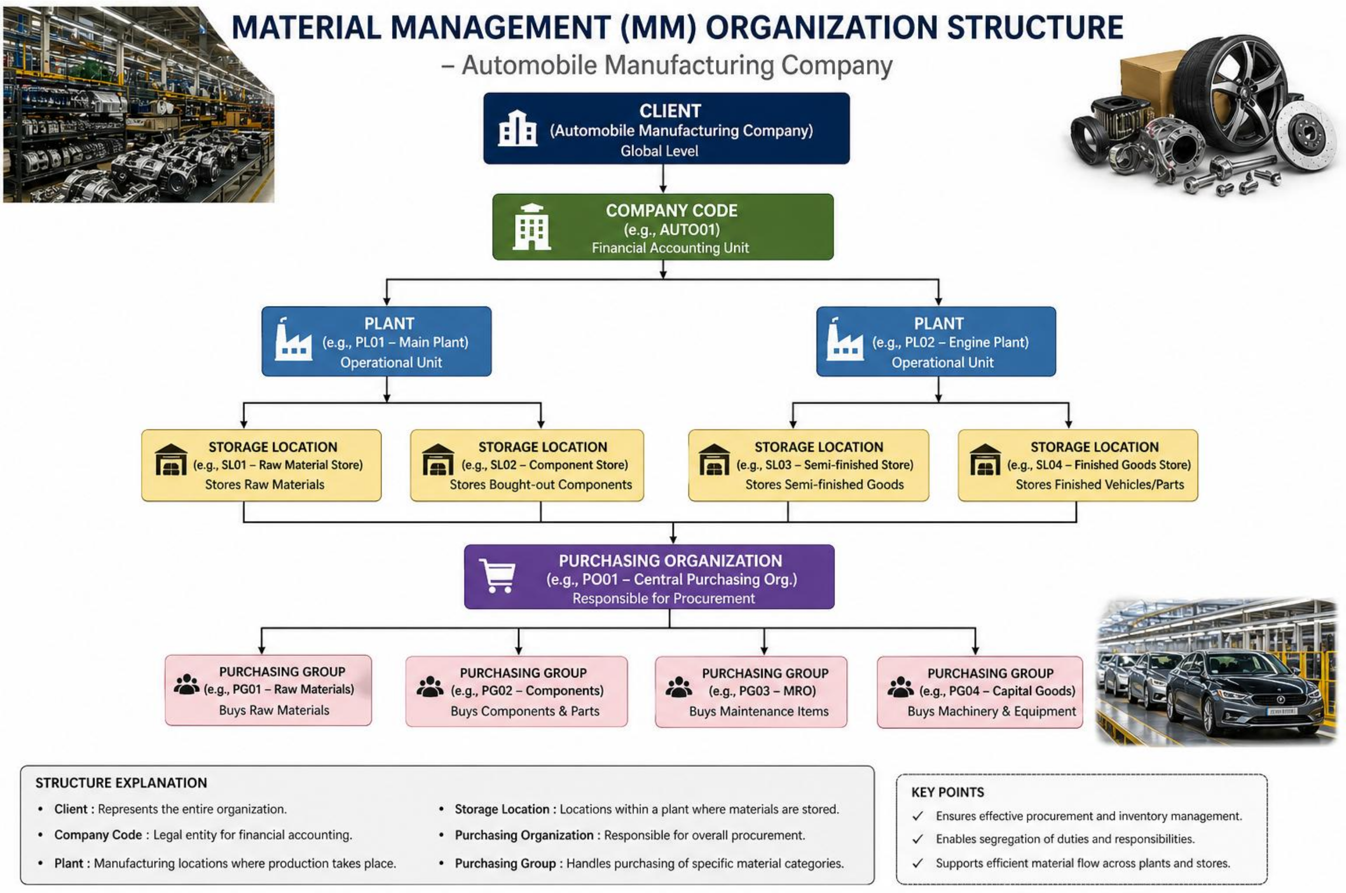
9. Organizational Structure

9.1 Overall Organization Structure



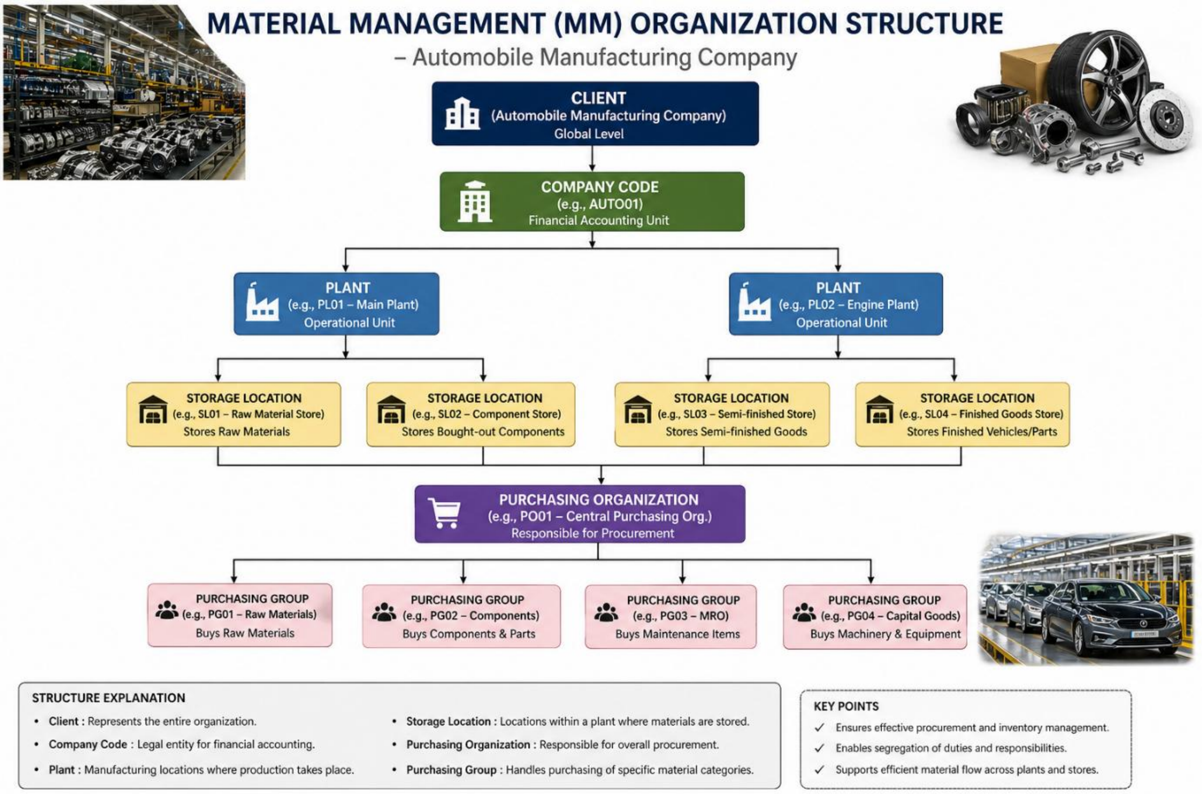
- The overall structure shows the hierarchy of the company.
- It includes top management and different departments like Production, Sales, and Finance.
- Each department has specific roles and responsibilities.
- It helps in smooth coordination and efficient decision-making.

9.2 Material Management Organization Structure



- The MM structure shows how materials are managed in the company.
- It includes Client, Company Code, Plant, and Storage Location.
- Purchasing Organization and Purchasing Group handle procurement.
- It helps in proper material flow and inventory control.

9.3 Sales and Distribution Organization Structure



- The SD structure shows how sales activities are managed.
- It includes Sales Organization, Distribution Channel, and Division.
- Sales Offices and Sales Groups handle different regions and customers.
- It helps in efficient sales and better customer service.

10. Master Data Required

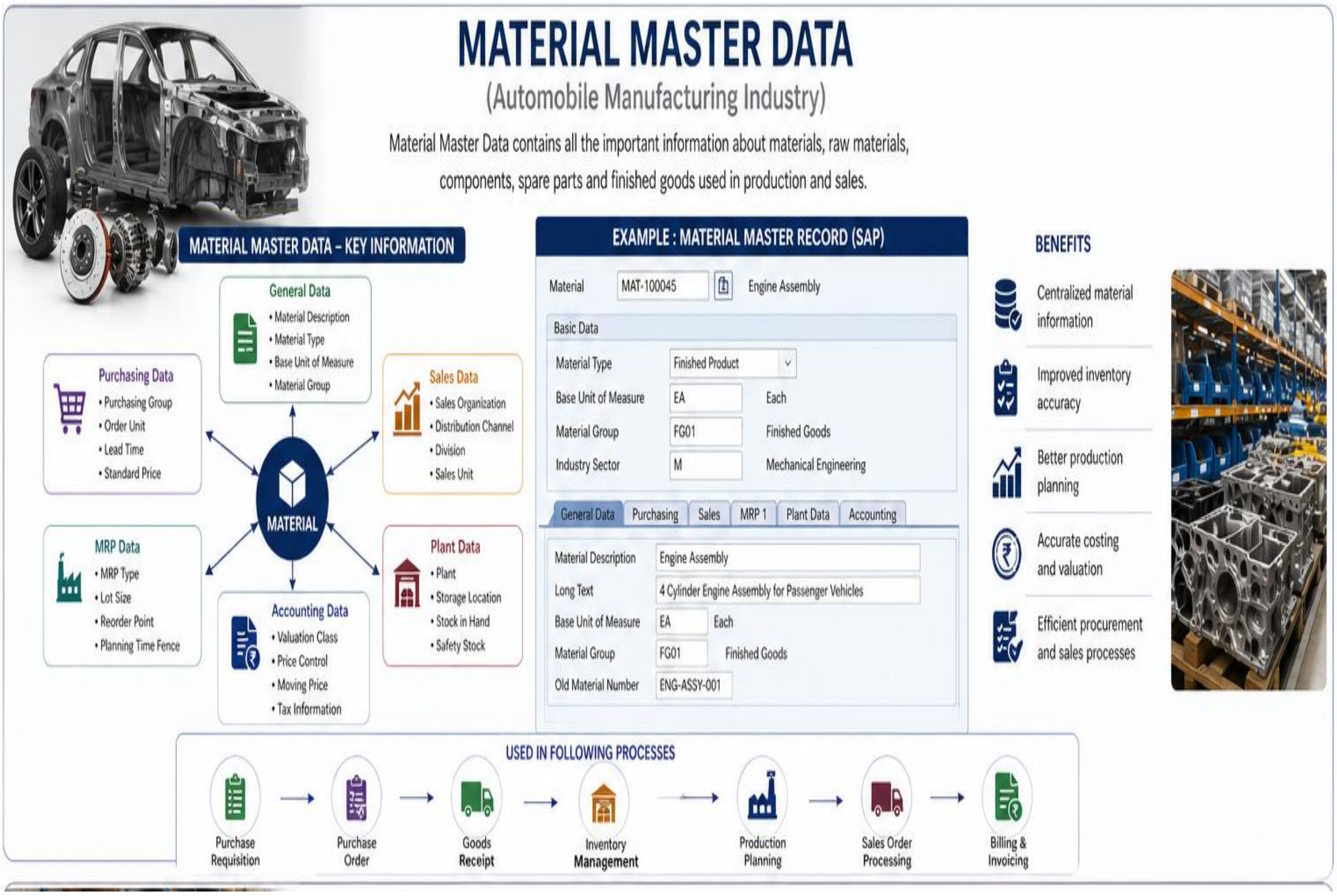
10.1 Customer Master

Customer Master contains all the important information related to customers who purchase vehicles or services from the automobile manufacturing company.

Key Details:

- It includes the customer’s name and address, which helps in identifying and locating the customer.
- It stores contact details such as phone number, email, and communication information for effective interaction.
- It maintains payment terms, including payment methods and conditions agreed with the customer.
- It contains sales data, such as sales organization, distribution channel, and customer-specific sales details.

10.2 Material Master



Material Master stores all the essential details of materials that are used in the production process of an automobile manufacturing company.

Examples:

- It includes engine components that are used in assembling vehicle engines.
- It contains tires that are required for vehicle assembly.
- It includes batteries that are used in automobiles.
- It also contains raw materials such as steel, plastic, and other basic inputs used in manufacturing.

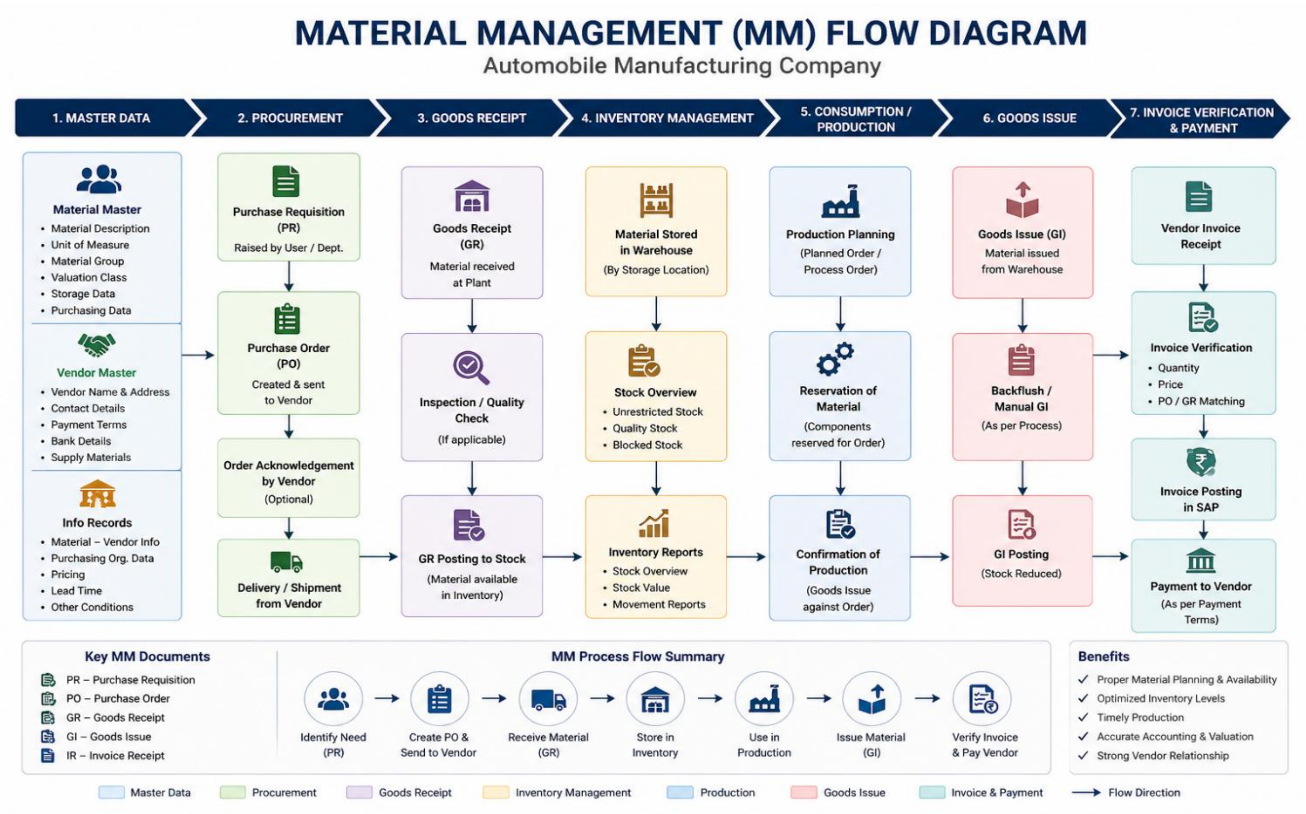
10.3 Vendor Master

Vendor Master contains all the important information related to suppliers who provide materials and services to the automobile manufacturing company.

Key Details:

- It includes the vendor’s name and address, which helps in identifying and locating the supplier.
- It stores contact information such as phone number, email, and communication details for effective coordination.
- It maintains payment terms, including payment methods and conditions agreed with the vendor.
- It records the materials or services supplied by the vendor to support procurement activities.

11.Material Management (MM):



Material Management (MM) in SAP is responsible for managing procurement, inventory, and material flow within the automobile manufacturing company.

1. Master Data Creation

The MM process begins with the creation of master data.

- Material Master is created to store details of materials such as raw materials, components, and finished goods.
- Vendor Master is created to maintain supplier information such as name, address, and payment details.

2. Purchase Requisition (PR)

- A purchase requisition is created internally to request the procurement of required materials.
- It includes details such as material type, quantity, and required delivery date.

3. Request for Quotation (RFQ)

- RFQs are sent to multiple vendors to obtain quotations.
- Vendors respond with pricing and delivery terms.

4. Quotation and Vendor Selection

- Quotations received from vendors are compared.

5. Purchase Order (PO) Creation

- A purchase order is created and sent to the selected vendor.
- It serves as a formal agreement for purchasing materials.

6. Goods Receipt (GR)

- Materials are received from the vendor at the plant or warehouse.
- A material document is generated to record the transaction.

7. Quality Inspection

- The received materials are inspected for quality (if required).
- Only approved materials are moved to unrestricted stock.

8. Inventory Management

- Materials are stored in appropriate storage locations.
- Inventory reports are generated for tracking and analysis.

9. Goods Issue (GI)

- Materials are issued from the warehouse for production.
- Stock levels are reduced accordingly.

10. Invoice Verification

- The vendor invoice is received and verified.
- It is matched with the purchase order and goods receipt (3-way matching).

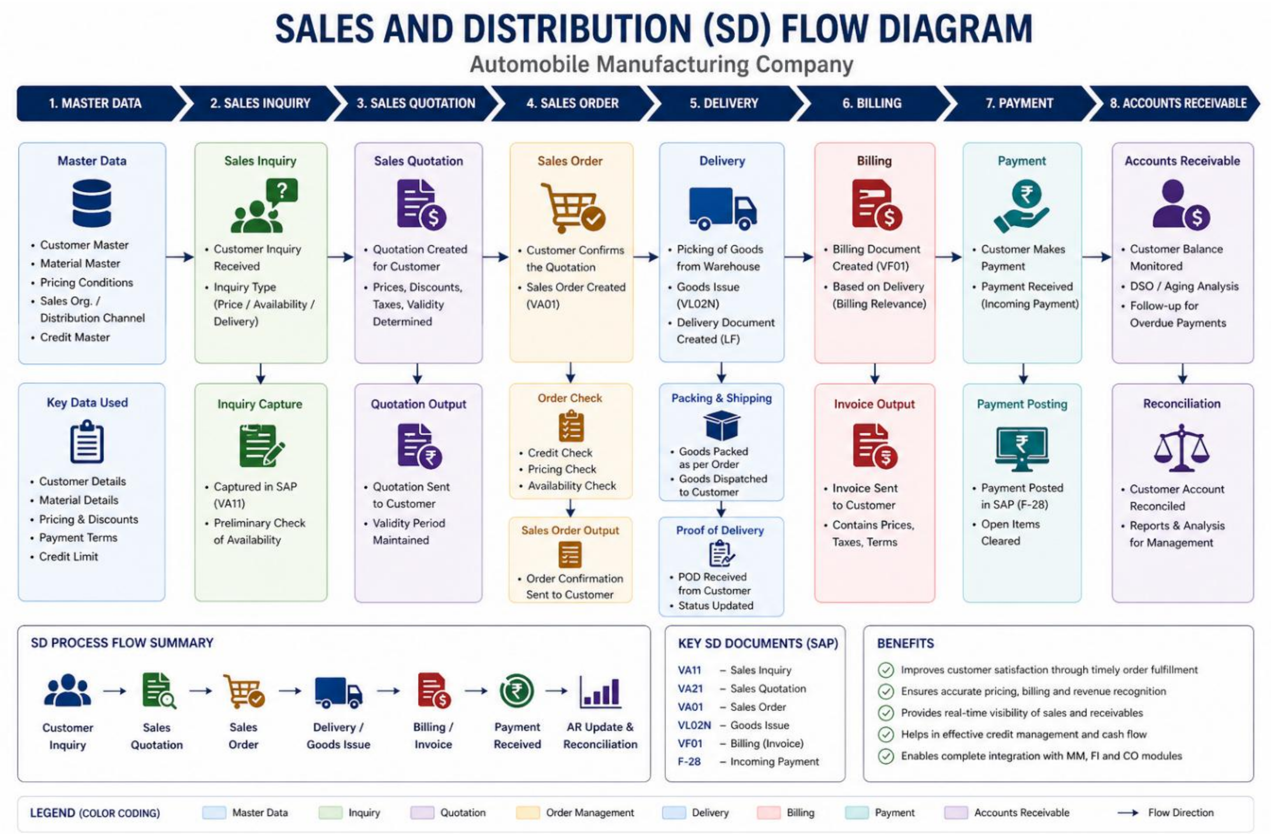
11. Payment Processing

- Payment is made to the vendor as per agreed terms.
- Financial records are updated in the system.

Key MM Documents

- Purchase Requisition (PR)
- Request for Quotation (RFQ)
- Purchase Order (PO)
- Goods Receipt (GR)
- Goods Issue (GI)
- Invoice Receipt (IR)

12.Sales and Distribution (SD):



Sales and Distribution (SD) in SAP manages all activities related to customer sales, order processing, delivery, billing, and payment. It ensures that customer requirements are fulfilled efficiently and that revenue is accurately recorded.

1. Master Data Creation

The SD process begins with maintaining master data.

- Customer Master is created to store customer details such as name, address, payment terms, and credit limits.
- Material Master is used to maintain product details such as price, availability, and specifications.

2. Sales Inquiry

- A sales inquiry is received from the customer regarding product availability, pricing, or delivery.
- It helps in understanding customer requirements.

3. Sales Quotation

- A quotation is created and sent to the customer.
- It includes pricing, discounts, taxes, and validity period.

4. Sales Order Creation

- Once the customer accepts the quotation, a sales order is created.
- It confirms the order details such as quantity, price, and delivery schedule.

5. Order Validation

- The system performs checks such as credit limit verification, pricing determination, and product availability.
- Ensures that the order can be processed without issues.

6. Delivery Processing

- Goods are picked from the warehouse.
- Packing and shipping activities are performed.
- Goods issue is posted, reducing inventory.

7. Billing

- An invoice is generated based on the delivery.
- Billing details include price, taxes, and payment terms.
- The invoice is posted to financial accounting.

8. Payment Processing

- Payment is received from the customer.
- The system records incoming payments and clears outstanding invoices.

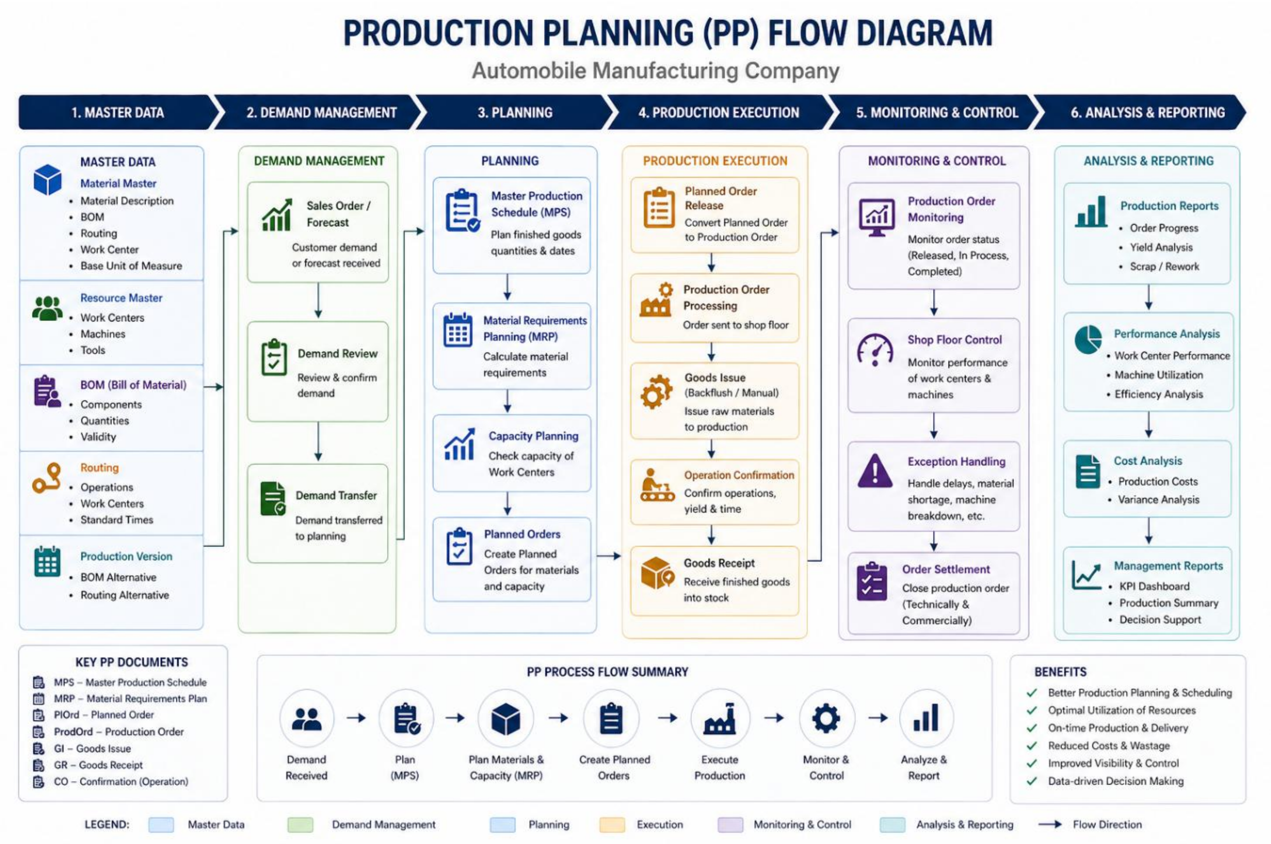
9. Accounts Receivable

- Customer account balances are monitored.
- Reports such as aging analysis and outstanding payments are generated.

Key SD Documents

- Sales Inquiry
- Sales Quotation
- Sales Order
- Delivery Document
- Billing Document
- Incoming Payment

13.Production Planning (PP):



Production Planning (PP) in SAP manages the planning, scheduling, and execution of production processes. It ensures that manufacturing is carried out efficiently based on demand and available resources.

1. Master Data Creation

The PP process starts with maintaining production-related master data.

- Material Master is maintained with production details.
- Bill of Materials (BOM) defines components required for production.
- Routing defines the sequence of operations and work centers.
- Work Centers represent machines or production units.

2. Demand Management

- Customer demand is received through sales orders or forecasts.
- Demand is reviewed and transferred to planning.

3. Production Planning

- Master Production Schedule (MPS) is created for finished goods.

- Material Requirements Planning (MRP) calculates required materials.
- Capacity planning ensures availability of machines and resources.
- Planned orders are generated for production.

4. Production Execution

- Planned orders are converted into production orders.
- Materials are issued to production.
- Production activities are carried out on the shop floor.
- Operations are confirmed after completion.
- Finished goods are received into inventory.

5. Monitoring and Control

- Production orders are monitored for progress and status.
- Shop floor activities are controlled for efficiency.
- Issues such as delays or shortages are handled.
- Production orders are settled after completion.

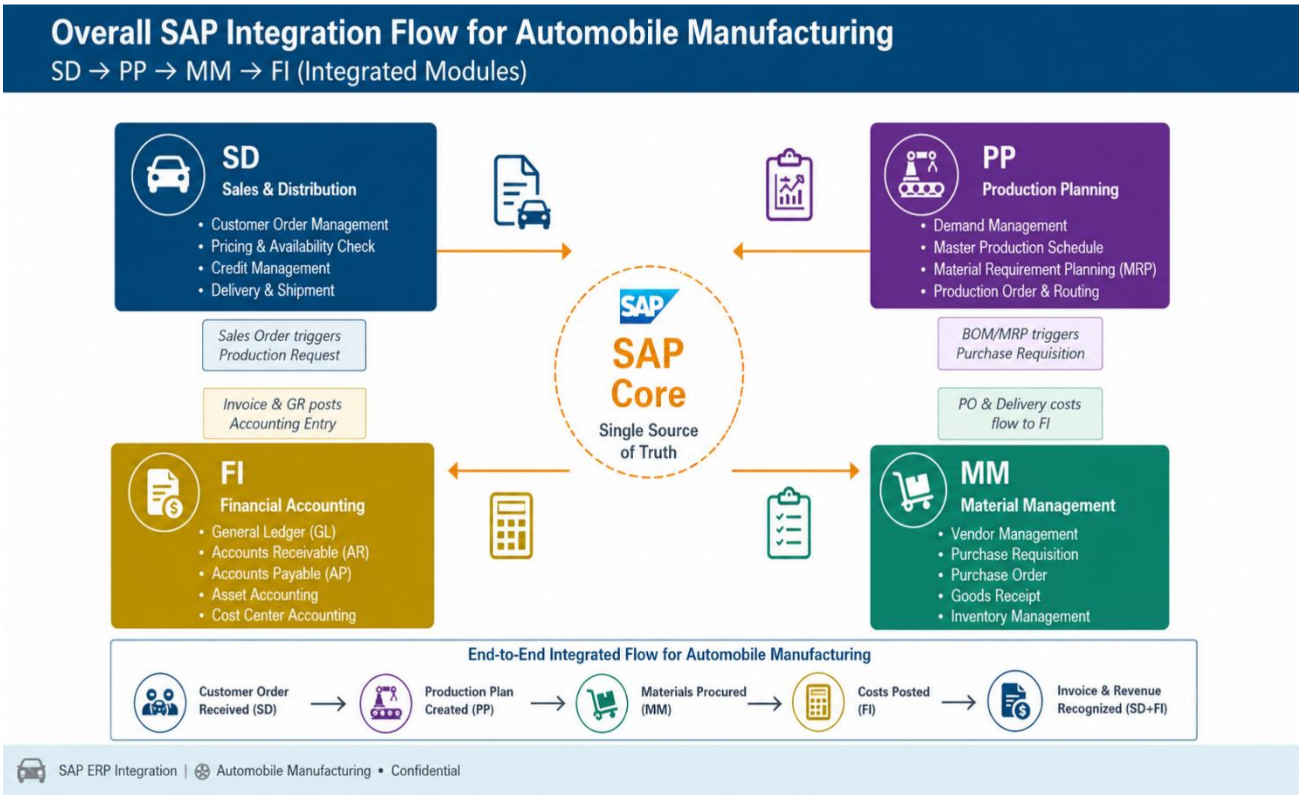
6. Analysis and Reporting

- Production reports are generated to track performance.
- Cost analysis is performed to evaluate production efficiency.
- Management reports support decision-making.

Key PP Documents

- Planned Order
- Production Order
- Goods Issue (GI)
- Goods Receipt (GR)
- Confirmation (CO)

14.Integration Between Modules (Very Important)



AP ERP integrates different modules such as Sales and Distribution (SD), Production Planning (PP), Material Management (MM), and Financial Accounting (FI) to ensure smooth and efficient business operations. This integration allows data to flow seamlessly across departments, providing a single source of truth.

Overall Integration Flow

The overall process in an automobile manufacturing company follows a structured sequence:

1. Customer order is received through the SD module
2. Production planning is carried out in the PP module
3. Materials are procured using the MM module
4. Financial transactions are recorded in the FI module

Module-wise Integration Explanation

1. Sales and Distribution (SD)

- The process starts when a customer places an order.
- The system performs pricing, availability checks, and credit verification.
- Once the sales order is confirmed, it triggers production planning in the PP module.

- After delivery and billing, invoice data is transferred to the FI module for accounting.

2. Production Planning (PP)

- Based on the sales order, demand is generated in the PP module.
- Production planning is carried out using MRP and scheduling.
- If required materials are not available, the system generates purchase requisitions.
- These requisitions are transferred to the MM module for procurement.

3. Material Management (MM)

- MM handles procurement of raw materials and components.
- Purchase requisitions are converted into purchase orders and sent to vendors.
- Goods are received and stored in inventory.
- The cost of procurement and inventory transactions is automatically updated in the FI module.

4. Financial Accounting (FI)

- FI records all financial transactions generated from SD, MM, and PP.
- It handles accounts receivable from customer billing.
- It manages accounts payable for vendor payments.
- It records production costs and maintains financial reports.

End-to-End Process Flow

The complete integrated flow can be summarized as follows:

- Customer order is received in SD
- Production plan is created in PP
- Materials are procured in MM
- Costs and accounting entries are recorded in FI
- Invoice is generated and revenue is recognized

Key Integration Points

- Sales Order in SD triggers demand in PP
- MRP in PP triggers procurement in MM
- Goods receipt and invoice in MM update FI
- Billing in SD updates accounts receivable in FI

15.Benefits of SAP ERP Implementation

The implementation of SAP ERP in an automobile manufacturing company offers numerous advantages that significantly enhance overall business performance:

- **Improved Efficiency and Productivity:**
  Automation of business processes reduces manual work, minimizes errors, and increases operational speed.
- **Real-Time Data Availability:**
  SAP ERP provides instant access to accurate and up-to-date information, enabling better monitoring and control of operations.
- **Better Inventory Management:**
  Efficient tracking of raw materials, work-in-progress, and finished goods helps in reducing excess inventory and avoiding stock shortages.
- **Faster Decision-Making:**
  With real-time insights and analytics, management can make quick and informed decisions.
- **Reduced Operational Costs:**
  Streamlined processes and better resource utilization lead to cost savings in production, procurement, and logistics.
- **Improved Customer Satisfaction:**
  Timely delivery, accurate order processing, and better service enhance customer experience and trust.
- **Integrated Business Processes:**
  Seamless integration of different departments such as procurement, production, sales, and finance ensures smooth information flow across the organization.

16.Challenges During SAP ERP Implementation

Despite its many benefits, implementing SAP ERP comes with certain challenges that organizations must carefully manage:

- **High Implementation Cost:**
  Initial investment in software, hardware, and consulting services can be significant.
- **Resistance to Change from Employees:**
  Employees may be reluctant to adopt new systems and processes, affecting implementation success.

- **Need for Training and Skill Development:**
  Proper training is essential to ensure employees can effectively use the system.
- **Data Migration Issues:**
  Transferring data from legacy systems to SAP ERP can be complex and error-prone if not handled carefully.
- **Complex System Configuration:**
  Customizing SAP to meet specific business requirements requires expertise and careful planning.
- **Time-Consuming Implementation Process:**
  Full implementation of SAP ERP can take several months or even years depending on the organization's size and complexity.

17.Conclusion

SAP ERP implementation in the automobile manufacturing industry provides a strong foundation for integrating business processes and improving operational efficiency. It enables organizations to streamline activities across various departments, ensuring better coordination and control.

The integration of key modules such as Material Management (MM), Sales and Distribution (SD), and Production Planning (PP) ensures smooth coordination between procurement, production, and sales functions. This integration plays a crucial role in maintaining a continuous and efficient production cycle.

With modern technologies like SAP Fiori, organizations can further enhance user experience, improve accessibility, and enable real-time decision-making through intuitive interfaces.

Overall, SAP ERP helps automobile manufacturing companies achieve operational excellence, optimize resource utilization, reduce costs, and maintain a competitive edge in the global market.

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SAP ERP Integration

in an Automobile Manufacturing Company

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Course: Enterprise Resource Planning (ERP)

Institution: SRM University AP

College Project Presentation

Introduction to ERP in the Automobile Industry

What is ERP? Enterprise Resource Planning (ERP) is integrated software that manages core business processes — finance, supply chain, production, sales — in real time on a single platform.

Why Automobile?

- High product complexity
- Global supply chains
- JIT manufacturing
- Strict quality norms
- Multi-location plants

ERP Benefits

- Real-time visibility
- Process automation
- Data accuracy
- Cost reduction
- Better decisions

Industry Scale

- 100s of suppliers
- Thousands of parts
- Crores in daily revenue
- Pan-India distribution
- Regulatory compliance

About the Company — Toyota Motor Corporation

TOYOTA

Toyota Motor Corporation (Real Case Study)

- **Established:** 1937
- **HQ:** Toyota City, Japan
- **Employees:** 370,000+ worldwide
- **Products:** Passenger cars, SUVs, hybrid vehicles, commercial vehicles
- **Global Presence:** 170+ countries

~10 Million Units / Year	10,000+ Suppliers	5,000+ Dealers	$250+ Billion Revenue
Global production	Global supply chain network	Worldwide	Annually (FY 2024)

Existing IT Landscape (Before SAP)

- Multiple legacy systems across regions
- Limited real-time supply chain visibility
- Manual coordination between plants & suppliers
- Disconnected procurement and finance systems
- Data silos across manufacturing & sales

> *Toyota partnered with SAP to build an integrated, intelligent enterprise for the future.*

SAP S/4HANA

Existing System & Key Challenges

Data Silos
Each department maintained separate data — no single source of truth

Manual Errors
Inventory counted manually; frequent mismatch in stock records

Delayed Reporting
MIS reports took 3-5 days; leadership had no real-time dashboards

Poor Coordination
Sales, production & procurement teams worked in isolation

Vendor Issues
No automated PO generation; 40% orders placed via phone/email

Compliance Risk
GST filing errors due to manual data entry across finance teams

Why SAP ERP? — The Right Choice

SAP

Systems, Applications & Products

SAP is the world's leading ERP vendor, trusted by 400,000+ customers across 180 countries. It is specifically designed for discrete and process manufacturing — making it the ideal fit for automobile companies.

01 — Industry Best Practices

Pre-built auto industry processes (ASAP methodology)

02 — Real-Time Integration

All modules talk to each other — one database, zero duplication

03 — Scalability

Handles growth from 1 plant to 20 plants seamlessly

04 — Compliance & GST

Built-in Indian tax engine for GST, TDS, statutory compliance

05 — Strong Analytics

SAP BW/BI for powerful dashboards & executive MIS

06 — Vendor Ecosystem

Largest partner network for support, customization, training

SAP Modules Overview — SD, MM, PP, FI

SD
Sales & Distribution

- Order to cash process
- Customer master & pricing
- Delivery & billing
- Sales reporting

MM
Material Management

- Procurement process
- Inventory management
- Vendor management
- Goods receipt & invoice

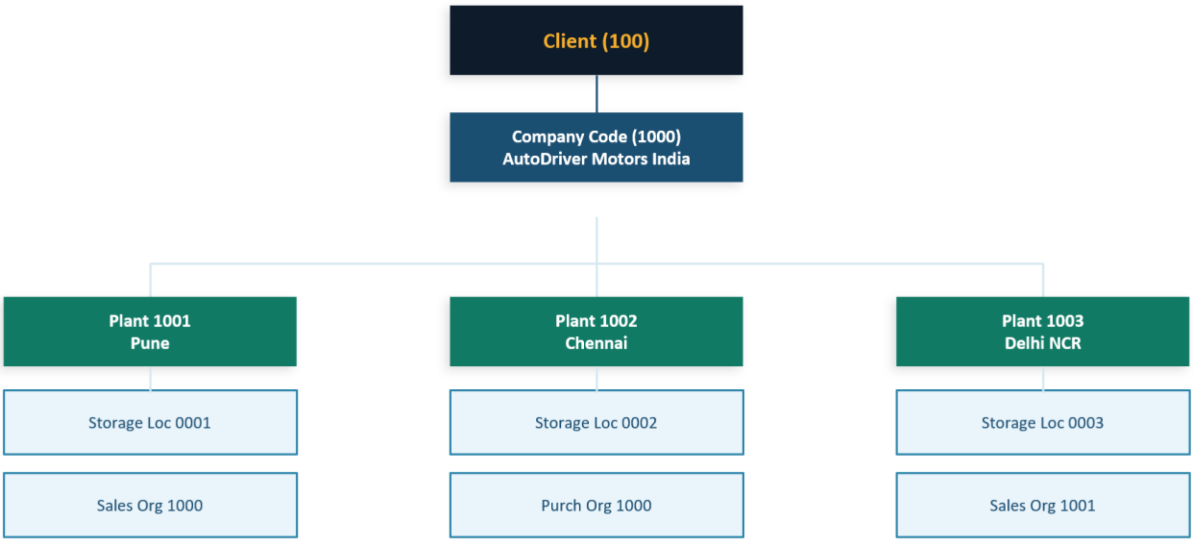
PP
Production Planning

- Production orders
- BOM & routing
- MRP planning
- Shop floor control

FI
Financial Accounting

- General ledger
- Accounts payable/receivable
- Asset accounting
- GST & tax compliance

SAP Organizational Structure



Key: Client → Company Code → Plant → Storage Location / Sales Org / Purchase Org

Master Data — Customer, Material & Vendor

Master data is the foundation of SAP — created once, used across all modules.

Customer Master (SD)

- Customer ID, Name & Address
- Sales organization & division
- Payment terms & credit limit
- Pricing conditions & discounts
- Delivery & shipping details

Material Master (MM/PP)

- Material number & description
- Material type (Finished/Raw/Semi)
- Unit of measure & weight
- MRP type & lot size
- Storage & purchasing views

Vendor Master (MM/FI)

- Vendor ID & company details
- Bank details for payment
- Payment terms & currency
- Purchasing organization link
- Reconciliation account (FI)

Material Management (MM) — Procurement Process

1	2	3	4	5	6
Purchase Requisition	**Request for Quotation (RFQ)**	**Purchase Order (PO)**	**Goods Receipt (GR)**	**Invoice Verification**	**Payment to Vendor**
PR auto-generated from MRP run	*RFQ sent to registered vendors*	*PO raised & approved in SAP*	*GR posted; stock updated*	*3-way match: PO-GR-Invoice*	*Auto-payment via FI module*

Key MM Configurations & Features

- Vendor evaluation with scoring matrix
- Automatic reorder point planning
- Material requirements planning (MRP)
- Batch management for part tracking
- GR/GI slips auto-printed
- Source list & quota arrangement

Sales & Distribution (SD) — Order to Cash

1	2	3	4	5	6
Inquiry / Quotation	**Sales Order (SO)**	**Delivery Creation**	**Goods Issue (GI)**	**Invoice / Billing**	**Payment Receipt**

SD Process — Key Steps

- Customer places order via dealer portal
- SO created with pricing, payment & delivery terms
- Credit check performed automatically
- Delivery document created from SO
- Goods Issue reduces inventory in real-time

SD Integration & FI Link

- Invoice generated automatically after GI
- GST calculated automatically in billing
- Customer account debited in FI module
- Returns & credit memo handled in SD
- Revenue recognized as per accounting rules

Production Planning (PP) — Make-to-Order Process

Sales Order → MRP Run → Planned Order → Production Order → Goods Issue → Production Confirm → Goods Receipt

Bill of Materials (BOM)

Complete list of components needed to build each vehicle model

MRP Planning

Auto-calculates raw material requirements based on demand

Routing

Step-by-step manufacturing operations with time standards

Capacity Planning

Checks plant capacity to avoid bottlenecks before production

Work Centers

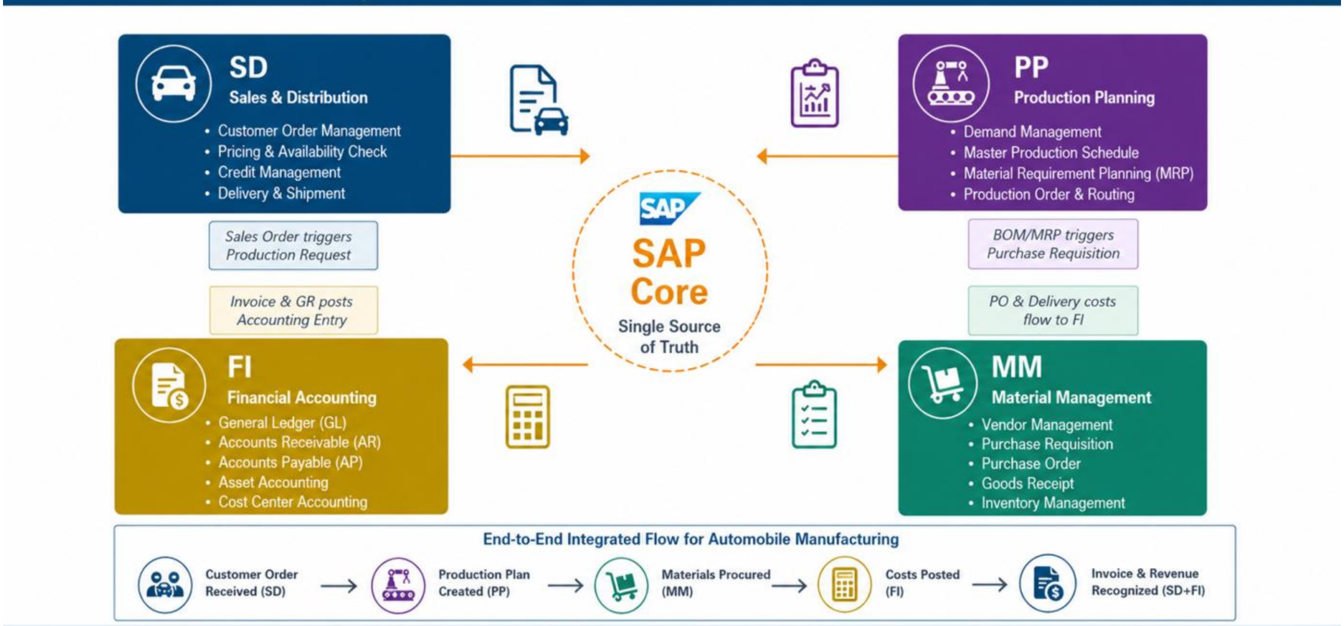
Machines/assembly lines defined with capacity & cost rates

Shop Floor Control

Real-time tracking of production order progress

Overall SAP Integration Flow for Automobile Manufacturing

SD → PP → MM → FI (Integrated Modules)



Benefits of SAP ERP Implementation

📊 Real-Time Visibility

Live dashboards for inventory, production & financials — decisions in minutes, not days

⚡ Process Automation

Purchase orders, invoices & MRP runs automated — 60% reduction in manual work

💰 Cost Reduction

Inventory carrying costs reduced by 25%; reduced excess procurement via MRP

🎯 Better Quality

Batch traceability & QM module ensures defect tracking from supplier to end-customer

📋 GST Compliance

Built-in tax engine handles GST, TDS, TCS — zero manual errors in statutory filings

🌐 Scalability

Easily add new plants, product lines or sales regions without system overhaul

Conclusion

Key Takeaway: SAP ERP integration transforms AutoDrive Motors from a siloed, manual operation into a data-driven, automated, and globally connected enterprise — driving efficiency, compliance, and competitive advantage.

Summary Points

- ✅ SAP consolidates SD, MM, PP, and FI into one unified platform
- ✅ Real-time integration eliminates data silos & reduces errors
- ✅ MRP-driven procurement ensures right material, right time, right quantity
- ✅ Automated billing and payment through FI improves cash flow
- ✅ GST-compliant system reduces regulatory risk
- ✅ Scalable architecture supports future business growth

Thank You!

SAP ERP Integration in an Automobile Manufacturing Company

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Open for Questions & Discussion